AERO60492 – Autonomous Mobile Robots
Coursework 3 – Feedback Control, lab experiment

Aims
The aim of the series of these labs is to test your feedback controllers in real world settings. You need to test the controller and report on its performance. Note that your controller is not expected to work (refer to markscheme in general coursework sheet), instead the focus is on your understanding why it did/didn't work and how would you improve.

Timeline (experiment)
Week 9 – Initial test of your controller
Week 10 – Final test of your controller
Note: there might be one more session which we are currently trying to schedule. This session is intended as emergency only. **Do not** expect/rely on it to happen.

Task: Test controller in experiment

The objective of the controller is to achieve the desired position and yaw by controlling x,y,z velocities and yaw rate. We will provide you with the current positions and attitudes of the quadcopter. The interface is exactly the same as in the simulator. Essentially, your task is to design an algorithm that takes the current state of the UAV and outputs velocity commands to guide and maintain the UAV at the desired position.

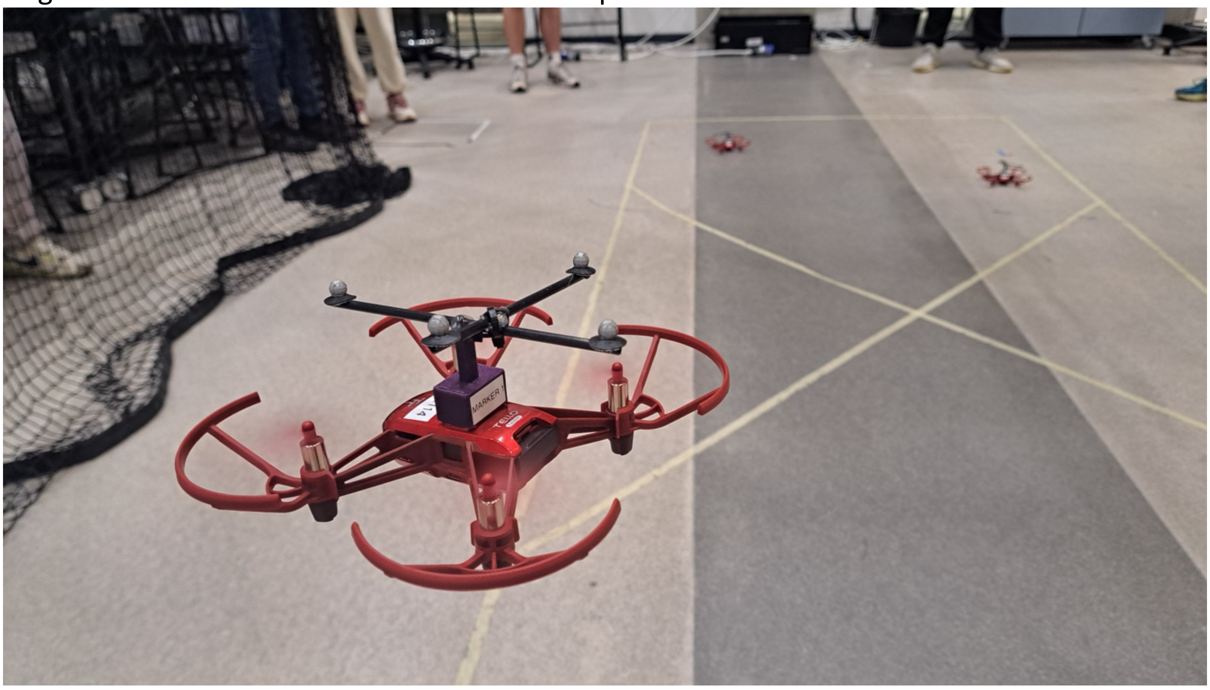

Figure 1. The tello drone used in the experiment

In this experiment we will be using a DJI tello quadcopter (Figure 1). The interface will be the same as for the simulated quadcopter. However, physics and behaviour of the tello will not be the same. You should not expect your controller to work out of the box. The experimental Tello will be limited to flying +/- 1 meter in each direction. It is up to the teams to record the videos and data of the tellos. Each team will be given one 30 minutes slot in each lab. You should have one laptop prepared to run your controller. You can find the instruction on blackboard.

NOTES (experiment only):

- What data you chose to save is down to discretion of each team. Make sure you save enough to be able to use it in the video. You are allowed to use whatever library you want for saving the data. Make sure to remove it/not use it for simulation code, otherwise the simulator will fail.